COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 1 : SLO 1 — State and Prove De Morgan's Theorems
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1. State De Morgan's Theorems (Remember)

• The complement of the sum of two variables is equal to the PRODUCT of the COMPLEMENTS of the variables. (A + B)' = A' . B'
• The complement of the product of two variables is equal to the SUM of the COMPLEMENTS of the variables. (A . B)' = A' + B'

2. Match the Expressions (Understand)

• 1. (X + Y)' → d. The complement of a sum
• 2. X' . Y' → c. The product of the complements
• 3. (X . Y)' → b. The complement of a product
• 4. X' + Y' → a. The sum of the complements

3. Complete the Proof (Apply) — Truth table for (A + B)' = A' . B'

• A=0, B=0: A+B=0, (A+B)'=1, A'=1, B'=1, A'.B'=1 → LHS = RHS
• A=0, B=1: A+B=1, (A+B)'=0, A'=1, B'=0, A'.B'=0 → LHS = RHS
• A=1, B=0: A+B=1, (A+B)'=0, A'=0, B'=1, A'.B'=0 → LHS = RHS
• A=1, B=1: A+B=1, (A+B)'=0, A'=0, B'=0, A'.B'=0 → LHS = RHS
• Since the columns for (A + B)' and A' . B' are identical in every row, the theorem is proved.

4. Simplify the Expression (Analyze)

• F = ((A . B)' + C)'
• Step 1: F = ((A . B)')' . C' (De Morgan's theorem applied to the outer complement)
• Step 2: F = (A . B) . C' (law of double complementation)
• Final simplified expression: F = A . B . C'

5. Construct a Proof (Create) — Truth table for (A . B)' = A' + B'

• A=0, B=0: A.B=0, (A.B)'=1, A'=1, B'=1, A'+B'=1 → LHS = RHS
• A=0, B=1: A.B=0, (A.B)'=1, A'=1, B'=0, A'+B'=1 → LHS = RHS
• A=1, B=0: A.B=0, (A.B)'=1, A'=0, B'=1, A'+B'=1 → LHS = RHS
• A=1, B=1: A.B=1, (A.B)'=0, A'=0, B'=0, A'+B'=0 → LHS = RHS
• The columns for (A . B)' and A' + B' are identical, so the second theorem is proved.